Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Dornhoffer, John M.D.		
Project Title:	Pharmacological Countermeasures for Space Motion Sickness		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dornhofferjohnl@uams.edu	**Fax:**	FY 501-686-8029
PI Organization Type:	UNIVERSITY	**Phone:**	501-686-5016
Organization Name:	University of Arkansas for Medical Sciences		
PI Address 1:	4301 West Markham		
PI Address 2:	MS-543		
PI Web Page:			
City:	Little Rock	**State:**	AR
Zip Code:	72205	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	03/01/2001	**End Date:**	06/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Garcia-Rill, Edgar (University of Arkansas for Medical Sciences) Paule, Merle (National Center for Toxicological Research) Van de Heyning, Paul (University Hosptial of Antwerp)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Aim 1 will determine the effects of drug countermeasures currently used to treat the vertiginous patient (lorazepam, meclizine, promethazine, scopolamine), in alleviating motion sickness induced by vestibular stimulation with a rotary chair. Aim 2 will determine the effects of these countermeasures on cognitive performance using an operant Test Battery to assess time perception, short-term memory, and learning, and measurment of the P50 potential to assess any deficits in senory gating. Aim 3 will use 3D oculography and unilateral otolith testing to determine the extent of correlation between vestibular dysfuntion induced by the roatry chair and unloading of otolithic organs due to 0G.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study addressed one of the main exploration-mission risk areas set forth by NASA in the Critical Path Roadmap (Impaired cognitive and/or physical performance due to motion sickness symptoms or treatments, especially during/after G-level changes [Risk Type III, Risk Rank 3]) and had a countermeasure readiness level of 6. We have addressed critical question 9.12: How effective are other drugs in providing fast relief in mission critical situations and does the drug have unacceptable side effects, particularly the short term effects on cognitive function? In addition, through the completion of this study, we have standardized measures of oculomotor function, postural stability, and cognitive performance. These standards are crucial for establishing the effectiveness and quantifying the side effects of potential drug countermeasures. During our 2-year grant period, we have obtained salient findings in completion of the objectives of the original proposal. The results of our study have clearly indicated scopolamine to be the countermeasure of choice, leading to an National Space Biomedical Research Institute (NSBRI) renewal proposal that will investigate optimization of the dose and delivery of this countermeasure. Thus, our renewal focuses on optimizing drug delivery parameters for scopolamine, testing potential combination therapies, and examining the feasibility of scopolamine as rescue therapy. The proposed experiments are crucial for addressing one of the key impediments to human spaceflight: namely, the need to retain cognitive function while treating Smith-Magenis Syndrome (SMS). Our studies should yield optimal therapy for SMS that will allow astronauts to avoid the nausea, vomiting, and dizziness of SMS while allowing them to meet the attentional, cognitive, and physical requirements of spaceflight. Our proposed study addresses one of the main exploration-mission risk areas set forth by NASA in the Critical Path Roadmap (Risk #35): Impaired cognitive and/or physical performance due to motion sickness symptoms or treatments. It represents a countermeasure readiness level of 6-7. The critical questions this proposed study would address are: How effective are other drugs in providing fast relief in mission critical situations? Does the drug have unacceptable side effects, particularly the short term effects on cognitive function? (9.12) What is the physiological basis of space motion sickness? (9.26) Because it is a clinical study, our results should be rapidly adaptable to space-based research trials. If our off-axis work reveals a correlation between inner ear dominance and resistance to SMS, future research will also focus on developing a reliable screening paradigm for astronauts using off-axis rotation. Thus, the subjective visual vertical (SVV)/rotational paradigm could be used to determine those individuals who would be most susceptible to SMS, and they would receive scopolamine pre-flight.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Paule MG, Chelonis JJ, Blake DJ, Dornhoffer JL. "Effects of drug countermeasures for space motion sickness on working memory in humans." Neurotoxicol Teratol. 2004 Nov-Dec;26(6):825-37. PMID: 15451046 , Nov-2004
Articles in Peer-reviewed Journals	Dornhoffer J, Chelonis JJ, Blake D. "Stimulation of the semicircular canals via the rotary chair as a means to test pharmacologic countermeasures for space motion sickness." Otol Neurotol. 2004 Sep;25(5):740-5. PMID: 15354005 , Sep-2004